Lucid: 13-Node LangGraph Multi-Agent Workflow

Five-Stage Sequential Research, Deterministic Optimization & Fiduciary Explanations

SHARED GRAPH STATE: PortfolioAgentState (35 Fields in backend/agent/state.py) Mandate (11) • Universe (3) • Analysts (4) • Critic (7) • Math & Proofs (10)

STAGE 1: SUITABILITY GATE

Node 1: Manager
Parse Mandate & Init State

Node 2: Suitability Checker
₹10k Min • 504 Liquid Equities

D1: Pass?
≥5 Stocks/3 Sec

Pass

Suitability Approved
504 Liquid Pool to Stage 2

STAGE 2: CHAINED SPECIALIST RESEARCH ANALYSTS

N3: Macro Analyst
RBI Repo 6.50% • CPI 5.10%
OUT: Cash Buffer 2.5–5%

N4: Market Analyst
India VIX • 200-DMA Trend
OUT: Regime & Beta Cap

N5: Fundamental Analyst
ROE ≥15% • D/E ≤1.0 • P/E
OUT: Winsorized Z-Scores

N6: News & Regulatory
30d News • Pledges <5%
OUT: Sentiment [-1, +1]

STAGE 3: STRATEGIST & CRITIC LOOP

Node 7: Lead Strategist
Selects K ∈ [10..16] Candidates
Core • Growth • Defensive

Rework (≤2)

Node 8: Adversarial Critic
Mandate & Concentration Audit
Temp 0.1 • Strict Skeptic

D2: Approved?
Caps & Ethics

Approved

Vetted Candidates
10–16 Critic-Approved Stocks
Dispatched to SLSQP Solver

STAGE 4: ZERO-LLM DETERMINISTIC MATH & VALIDATION

Node 10A: SciPy SLSQP
Max Sharpe / Min Variance
Bounds: Stock ≤25%, Sec ≤40%

Node 10B: Integer Sizing
Floor Lots: $Q_i = \lfloor (w_i^* \cdot V)/P_i \rfloor$
Guaranteed Cash Reserve ≥ ₹0

D4: Valid?
Stock/Sec Caps

Pass

Node 11: Compliance Gate
100% Invariants Satisfied
Div Score 95.5 • Sharpe 0.41

STAGE 5: FIDUCIARY EXPLANATION & PERSISTENCE

Node 12: Fiduciary Moderator
Level-1 Thesis & Level-2 NSE/BSE Proofs

Node 13: PostgreSQL ACID Commit
Atomic Write: 6 Tables (Versions, Decisions)

ACTIVE PORTFOLIO
Whole Shares Ready for Review

ARCHITECTURAL PILLAR: Strict Separation of Qualitative AI (Nodes 3-9, 12) from Deterministic Math (Nodes 10-11) — Zero Hallucinations